Code

Stats 20, F23 – Homework 4

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Question 1

```
my_t <- function(x) {
  new_matrix <- matrix(nrow = ncol(x), ncol = nrow(x))
  for (i in 1:nrow(x)) {
    for (j in 1:ncol(x)) {
      new_matrix[j,i] <- x[i,j]
    }
  }
  new_matrix
}
```

.

Question 2

(a)

```
my_cor <- function(x, y, use = "everything") {
  xbar <- mean(x)
  ybar <- mean(y)
  diffx <- x-xbar
  diffy <- y-ybar
  diffx_2 <- (x-xbar)^2
  diffy_2 <- (y-ybar)^2

#  if (use == "everything") {
#    sum(diffx * diffy) / (sqrt(sum(diffx_2)) * sqrt(sum(diffy_2)))
#  }

  if (use == "pairwise.complete.obs") {
    x = x[!is.na(x) & !is.na(y)]
    y = y[!is.na(x) & !is.na(y)]

  }

  sum(diffx * diffy) / (sqrt(sum(diffx_2)) * sqrt(sum(diffy_2)))

}
```

```
x <- c(1, 2, 3, 4, 5)
y <- c(2, 3, 4, 5, 6)
test <- my_cor(x,y,use = "pairwise.complete.obs")
test
```

```
## [1] 1
```

```
real <- cor(x,y,use = "pairwise.complete.obs")
real
```

```
## [1] 1
```

(b)

```
linreg <- function(x,y,use = "everything"){

  if (use == "pairwise.complete.obs") {
    x <- x[!is.na(x) & !is.na(y)]
    y <- y[!is.na(x) & !is.na(y)]
  }

  b = my_cor(x,y) * (sd(y) / sd(x))
  a = mean(y) - (b * mean(x))
  c(a,b)
}
```

(c)

```
height <- c(61,62,63,64,66,68)
weight <- c(104,110,125,141,160,170)
linreg(height,weight)
```

```
## [1] -510.64706   10.08824
```

The equation would be y = -510.64706 + 10.08824x

(d)

```
linreg_mat <- function(x,y) {
  if (length(x) != length(y)){
    stop("Vector arguments must be of same length.")
  }
  X = matrix(data = c(rep(1,length(x)), x), nrow = length(x),ncol = 2)
  Xt <- my_t(X)
  ((Xt %*% X)^-1) %*% Xt %*% y
  solve
  # figure out hint 3
}
```

(e)

```
linreg_mat(height,weight)
```

```
## function (a, b, ...)
## UseMethod("solve")
## <bytecode: 0x15a7b45b8>
## <environment: namespace:base>
```

My answers are very different from those in part (c), -510 vs 270 and 10 vs 4.

(f)

The slope coefficient means that for every unit change in our independent variable (x), height, we predict a 4.22 unit increase in weight.

Question 3

(a)

```
`%m%` <- function(A,B) {
  m = nrow(A)
  n = ncol(A)
  p = ncol(B)

  C = matrix(0, ncol = p, nrow = m)
  if (ncol(A) == nrow(B)) {
    for (i in 1:m){
      for (j in 1:p){
        for (k in 1:n) {
          C[i,j] <- C[i,j] + (A[i, k] * B[k, j])
        }
      }
    }
    C
  }
  else {
    stop("Matrix arguments are not conformable.")
  }
}
```

(b)

```
X <- matrix(c(6,3,5,2,4,1), nrow = 2, ncol = 3)
Y <- matrix(c(1:9), nrow = 3, ncol = 3)
mine <- X %m% Y
built <- X %*% Y
mine
```

```
##      [,1] [,2] [,3]
## [1,]   28   73  118
## [2,]   10   28   46
```

```
built
```

```
##      [,1] [,2] [,3]
## [1,]   28   73  118
## [2,]   10   28   46
```

Question 4

(a)

```
`%^%` <- function(A,k) {
  result = A
  num = 1
  while (num < k) {
    result = result %m% A
    num = num + 1
  }
  result
}
```

(b)

```
Z <- matrix(c(0.2,0.6,0.4,0.7,0.2,0.1,0.1,0.2,0.5), nrow = 3, ncol =
3)
Z %^% 0
```

```
##      [,1] [,2] [,3]
## [1,]  0.2  0.7  0.1
## [2,]  0.6  0.2  0.2
## [3,]  0.4  0.1  0.5
```

```
Z %^% 5
```

```
##         [,1]    [,2]    [,3]
## [1,] 0.38870 0.38429 0.22701
## [2,] 0.40232 0.36828 0.22940
## [3,] 0.39754 0.36995 0.23251
```

```
Z %^% 50
```

```
##           [,1]  [,2]      [,3]
## [1,] 0.3958333 0.375 0.2291667
## [2,] 0.3958333 0.375 0.2291667
## [3,] 0.3958333 0.375 0.2291667
```

```
Z %^% 500
```

```
##            [,1]  [,2]      [,3]
## [1,] 0.3958333 0.375 0.2291667
## [2,] 0.3958333 0.375 0.2291667
## [3,] 0.3958333 0.375 0.2291667
```

Question 5

```
my_which <- function(x, arr.ind = FALSE) {
  if (arr.ind) {
    ind1 = 1
    result = matrix(nrow = sum(x), ncol = 2)
    for (i in 1:nrow(x)){
      for (j in 1:ncol(x)){
        if (x[i,j]) {
          result[ind1, 1] <- i
          result[ind1, 2] <- j
          ind1 = ind1 + 1
        }
      }
    }
  } else {
    index = 1
    result = c()
    for (i in seq_along(x)) {
      if (x[i]) {
        result[index] <- i
        index = index + 1
      }
    }
  }
  result
}
```

Question 6

```
load("mlb.RData")
```

(a)

(i)

```
load("mlb.RData")
tapply(hit, team, max)
```

```
## ARI ATL BAL BOS CHA CHN CIN CLE COL DET HOU KCA LAA LAN MIA MIL MIN
NYA NYN OAK
## 172 191 163 188 145 176 182 183 182 185 170 192 162 145 165 187 161
170 142 161
## PHI PIT SDN SEA SFN SLN TBA TEX TOR WAS
## 153 155 155 178 135 163 148 148 129 180
```

(ii)

```
tapply(hit, team, length)
```

```
## ARI ATL BAL BOS CHA CHN CIN CLE COL DET HOU KCA LAA LAN MIA MIL MIN
NYA NYN OAK
##  49  59  56  44  51  50  53  49  41  49  41  49  60  52  51  53
54  50  56  53
## PHI PIT SDN SEA SFN SLN TBA TEX TOR WAS
##  48  48  49  53  48  49  54  50  63  53
```

(iii)

```
tapply(hr>0, team, sum)
```

```
## ARI ATL BAL BOS CHA CHN CIN CLE COL DET HOU KCA LAA LAN MIA MIL MIN
NYA NYN OAK
##  18  17  22  18  18  16  23  21  19  18  17  19  23  17  16  23
18  19  19  16
## PHI PIT SDN SEA SFN SLN TBA TEX TOR WAS
##  23  18  19  17  19  21  24  17  21  19
```

(b)

(i)

```
tapply(hit[ab >100]/ab[ab>100], team[ab > 100], max)
```

```
##       ARI       ATL       BAL       BOS       CHA       CHN
CIN       CLE
## 0.2928571 0.3090615 0.3150685 0.3461538 0.2923977 0.3054945
0.3099315 0.3119266
##       COL       DET       HOU       KCA       LAA       LAN
MIA       MIL
## 0.2966102 0.2985075 0.3164794 0.3067227 0.3121019 0.3123288
0.2782462 0.3257840
##       MIN       NYA       NYN       OAK       PHI       PIT
SDN       SEA
## 0.2880795 0.3333333 0.3288889 0.2960526 0.2702079 0.2996032
0.2796935 0.3037543
##       SFN       SLN       TBA       TEX       TOR       WAS
## 0.2839196 0.3052434 0.3426573 0.2725173 0.2811245 0.3081285
```

(ii)

```
tapply(hit, team, sum) / tapply(ab, team, sum)
```

```
##       ARI       ATL       BAL       BOS       CHA       CHN
CIN       CLE
## 0.2349817 0.2567180 0.2391502 0.2683621 0.2411733 0.2583570
0.2537961 0.2586238
##       COL       DET       HOU       KCA       LAA       LAN
MIA       MIL
## 0.2559105 0.2413542 0.2549056 0.2452316 0.2417763 0.2501795
0.2374271 0.2522555
##       MIN       NYA       NYN       OAK       PHI       PIT
SDN       SEA
## 0.2497286 0.2491387 0.2344550 0.2521957 0.2343289 0.2535341
0.2349617 0.2543080
##       SFN       SLN       TBA       TEX       TOR       WAS
## 0.2389460 0.2489996 0.2584475 0.2398680 0.2439292 0.2541236
```

(c)

```
tapply(hr, list(pos,league), mean)
```

```
##                     AL         NL
## catcher      4.46153846  4.7586207
## centerfield  6.76315789  7.3939394
## firstbase   11.15217391 12.9142857
## leftfield    8.80434783  9.6808511
## pitcher      0.05152225  0.0547619
## rightfield  11.73170732  7.5348837
## secondbase   6.25000000  6.2800000
## shortstop   10.12903226  8.5517241
## thirdbase    9.02631579  7.9318182
```

Rightfield has the biggest difference between leagues.

(d)

```
median(tapply(hr, list(pos,team), length))
```

```
## [1] 3
```

Question 7

(a)

```
my_row <- function(x) {
    output <- matrix(nrow = nrow(x), ncol = ncol(x))
    for (i in 1:nrow(x)){
      for (j in 1:ncol(x)){
        output[i,j] <- i
      }
    }
    output
}
```

(b)

```
my_col <- function(x) {
    output <- matrix(nrow = nrow(x), ncol = ncol(x))
    for (i in 1:nrow(x)){
      for (j in 1:ncol(x)){
        output[i,j] <- j
      }
    }
    output
}

my_row(Z) == my_col(Z)
```

```
##       [,1]  [,2]  [,3]
## [1,]  TRUE FALSE FALSE
## [2,] FALSE  TRUE FALSE
## [3,] FALSE FALSE  TRUE
```

(c)

```
my_diag <- function(x,nrow = nrow(x),ncol = nrow(x)) {
  output = c(sum((my_row(x) == my_col(x))))
  index = 1
  for (i in 1:nrow(x)) {
    if (my_row(x) == my_col(x)){
      output[index] <- x[i,i]
      index = index + 1
    }
  }
  output
}
```